

Chapter 15 - The SID Family

The SID (Sound Interface Device) is the MOS 6581/8580 chip that shipped in the Commodore 64. Intuition Engine has **three** SID instances available simultaneously, named SID, SID2, and SID3. Each one is a complete SID with three voices, a filter, and a master volume. Together they give the IE nine SID voices, matching the conventions of 6SID-style C64 demo music.

The IE adds a small extension called **SID+** that widens the chip's 4-bit volume registers to 8 bits and lets a program override the C64's master volume with a per-voice level. SID+ is off by default; with it off, the chip is bit-exact compatible with both the 6581 and the 8580.

15.1 What the SID can show

Item	Value
Voices per chip	3, each with its own oscillator, envelope, control
Waveforms	Triangle, sawtooth, pulse (with PWM), noise
Filter	Low-pass / high-pass / band-pass with resonance
Envelope	ADSR, exponential (SID-style)
Modulation	Hard sync and ring modulation with the previous voice
Master volume	4-bit, 8-bit in SID+ mode
Chips available	3 independent SIDs (SID, SID2, SID3)

15.2 The register block

Each SID is 0x1D bytes wide. The three SIDs are mapped at:

Chip	Base	End
SID	0xF0E00	0xF0E1C
SID2	0xF0E30	0xF0E4C
SID3	0xF0E50	0xF0E6C

The register layout below is for SID; SID2 and SID3 use the same offsets relative to their bases.

15.2.1 Voice registers

Each voice has seven registers; voice n ($n = 1, 2, 3$) lives at offset $(n - 1) * 7$:

Offset	Name	Purpose
0x00	FREQ_L0	Frequency low byte
0x01	FREQ_HI	Frequency high byte
0x02	PW_L0	Pulse width low byte
0x03	PW_HI	Pulse width high (low nibble only)

Offset	Name	Purpose
0x04	CTRL	Control byte (see below)
0x05	AD	Attack / decay (high / low nibble)
0x06	SR	Sustain / release

CTRL bits:

Bit	Name	Meaning
0	GATE	1 starts the envelope; 0 releases it.
1	SYNC	Hard-sync to the previous voice.
2	RINGMOD	Ring-modulate with the previous voice.
3	TEST	Reset oscillator.
4	TRIANGLE	Enable triangle waveform.
5	SAWT00TH	Enable sawtooth waveform.
6	PULSE	Enable pulse waveform (uses PW).
7	NOISE	Enable noise waveform.

The waveform bits OR together: setting more than one combines them, as on the real chip.

15.2.2 Filter and master volume

Address	Name	Purpose
0xF0E15	FC_LO	Filter cutoff low (3 bits).
0xF0E16	FC_HI	Filter cutoff high byte.
0xF0E17	RES_FILT	Resonance (bits 4–7) and per-voice routing (bits 0–3).
0xF0E18	MODE_VOL	Master volume (bits 0–3), filter mode (bits 4–6), voice-3 mute (bit 7).
0xF0E19	SID_PLUS_CTRL	Bit 0 = SID+ mode enable.

RES_FILT routing bits:

Bit	Field	Meaning
0	FILT_V1	Route voice 1 through the filter.
1	FILT_V2	Route voice 2 through the filter.
2	FILT_V3	Route voice 3 through the filter.
3	FILT_EXT	Route external input through the filter.
4–7	RES	Filter resonance, 0–15.

MODE_VOL bits 4–7:

Bit	Field	Meaning
4	LP	Low-pass enabled.
5	BP	Band-pass enabled.

Bit	Field	Meaning
6	HP	High-pass enabled.
7	30FF	Voice 3 mute.

15.2.3 Read-only registers

The original SID exposes potentiometer, oscillator-3, and envelope-3 read-back registers at offsets 0x19–0x1C. These addresses also overlap with the IE-specific player registers listed in §15.3.

15.3 The C64-style player

Each SID instance has a small player that streams C64-style SID data into the chip. The block lives just after the read-only registers of the primary SID:

Address	Name	Purpose
0xF0E20	SID_PLAY_PTR	Address of the music data.
0xF0E24	SID_PLAY_LEN	Length in bytes.
0xF0E28	SID_PLAY_CTRL	1 = play, 2 = stop.
0xF0E2C	SID_PLAY_STATUS	Read-only status.
0xF0E2D	SID_SUBSONG	Subsong number (0–255).

15.4 Chip model

Two SID variants are available: SID_MODEL_6581 (the original, warmer chip with a non-linear filter) and SID_MODEL_8580 (the revised chip with a cleaner filter). The IE selects between them through internal configuration; their externally visible registers are identical.

15.5 SID+ extension

Writing 1 to SID_PLUS_CTRL enables **SID+**:

- Master volume becomes 8-bit (MODE_VOL bits 0–3 map to a wider range).
- Each voice gains an independent volume that overrides the master.
- The chip otherwise behaves as a standard SID.

Writing 0 returns to bit-exact SID behaviour.

15.6 BASIC keywords

Form	Effect
SID VOICE v , $freq$, pw , $ctrl$, ad , sr	Program one voice (v is 1–3).
SID VOLUME $level$	Set master volume (0–15).
SID FILTER $cutoff$, $resonance$, $routing$, $mode$	Configure the filter.

Form	Effect
SID PLUS ON / SID PLUS OFF	Enable / disable SID+.
SID PLAY <code>addr[, len[, subsong]]</code>	Start hardware-mode SID playback.
SID STOP	Stop SID playback.

A BASIC fragment that plays a sustained C-major triad on the primary SID:

```
10 POKE &H000F0800, 1           : REM AUDIO_CTRL = on
20 SID VOLUME 15
30 SID VOICE 1, 4291, 0, &H41, &H00, &HF0 : REM C, pulse + gate
40 SID VOICE 2, 5407, 0, &H41, &H00, &HF0 : REM E
50 SID VOICE 3, 6430, 0, &H41, &H00, &HF0 : REM G
```

15.7 Putting it together

A SID program runs at three levels. From BASIC: drive a single SID with SID VOICE/FILTER/VOLUME. From machine language: address SID, SID2, and SID3 directly through their MMIO blocks for full nine-voice arrangements. For existing C64 music files, point SID_PLAY_PTR/LEN at the data and write 1 to SID_PLAY_CTRL.

The next chapter covers TED audio, the C16's two-voice sibling.